# Trigonometry

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## Cosine and sine

**Definition 0.1** (Unit circle)**.** The **unit circle** [1] is the set of points $(x_1, x_2)$ in the plane with $x_1^2 + x_2^2 = 1$.

**Example 0.1** (Points on the unit circle)**.**

- $(1, 0)$ is on the unit circle, because $1^2 + 0^2 = 1$.
- $(0, -1)$ is on the unit circle, because $0^2 + (-1)^2 = 1$.
- $(\frac{3}{5}, \frac{4}{5})$ is on the unit circle, because $\frac{9}{25} + \frac{16}{25} = 1$.
- $(1, 1)$ is not on the unit circle, because $1^2 + 1^2 = 2 \neq 1$.

**Definition 0.2** (The number $\pi$)**.** The number $\pi \approx 3.14159$ [2] is half of the circumference of the unit circle (Definition 0.1).

**Example 0.2** (Distances around the unit circle)**.** The whole unit circle has circumference $2\pi$. Moving from $(1, 0)$ to $(-1, 0)$ along the upper half of the circle covers a distance of $\pi$, and moving from $(1, 0)$ to $(0, 1)$ covers a quarter of the circle, a distance of $\pi/2$.

**Definition 0.3** (Angle on the unit circle)**.** The **angle** between two points of the unit circle is how far apart they are along the circle, going the short way round. Let the counterclockwise distance from the first point to the second be $s \in [0, 2\pi)$. Then the angle is

$$\theta \stackrel{\text{def}}{=} \min\{s, 2\pi - s\}.$$

In words, the angle is the smaller of two distances from the first point to the second: the counterclockwise one, $s$, or the clockwise one, $2\pi - s$.

**Example 0.3** (Three angles on the unit circle)**.**

- The points $(1, 0)$, $(0, 1)$ are a quarter of the circle apart, so their angle is $\pi/2$.
- The points $(1, 0)$, $(-1, 0)$ are half of the circle apart, so their angle is $\pi$.
- The point $(1, 0)$ is 0 apart from itself, so their angle is 0.

**Definition 0.4** (Radian)**.** A **radian** is the angle on the unit circle between two points that are a distance of 1 apart along the circle. An angle of $\theta$ is then $\theta$ radians.

**Example 0.4** (Angles in radians)**.**

- The points $(0, 1)$, $(-1, 0)$ are a quarter of the unit circle apart, so their angle is $\pi/2$ radians.
- The points $(0, 1)$, $(0, -1)$ are half of the circle apart, so their angle is $\pi$ radians.

**Proposition 0.1** (Range of an angle on the unit circle)**.** *An angle between two points of the unit circle is at least* 0*, at most* $\pi$ *(Definition 0.2).*

Proof

*Proof.* The whole circle has length $2\pi$ (Definition 0.2). Going the short way round covers at most half of the circle, which is $\pi$. A distance along the circle is at least 0. □

**Definition 0.5** (Cosine)**.** For a real number $\theta$, start at the point $(1, 0)$ of the unit circle (Definition 0.1) and move a distance $|\theta|$ along the circle, counterclockwise if $\theta \geq 0$ and clockwise if $\theta < 0$. The first coordinate of the point reached is the **cosine** of $\theta$ [3], written $\cos\theta$.

**Example 0.5** (Cosines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\cos 0 = 1$.
- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\cos\pi = -1$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\cos(\pi/2) = 0$.
- $\theta = \pi/4$ goes an eighth of the way round, to the point of the circle with $x_1 = x_2 > 0$. There $x_1^2 + x_1^2 = 1$, so $x_1^2 = \frac{1}{2}$ and $\cos(\pi/4) = x_1 = \frac{1}{\sqrt{2}} \approx 0.7071$.

Angles are also measured in *degrees*, with a full turn of $2\pi$ equal to 360 degrees, so $\pi/2$ is 90 degrees and $\pi/4$ is 45 degrees.

**Definition 0.6** (Sine)**.** For a real number $\theta$, the second coordinate of the point reached from $(1, 0)$ by moving a distance $|\theta|$ along the unit circle (Definition 0.5) is the **sine** of $\theta$ [3], written $\sin\theta$. The sine is one of the trigonometric functions [4].

**Example 0.6** (Sines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\sin 0 = 0$.
- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\sin\pi = 0$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\sin(\pi/2) = 1$.
- $\theta = \pi/4$ reaches the point with $x_1 = x_2 = \frac{1}{\sqrt{2}}$ (Example 0.5), so $\sin(\pi/4) = \frac{1}{\sqrt{2}}$.

**Theorem 0.1** (The Pythagorean identity)**.** *For every real number $\theta$ [5],*

$$\cos^2\theta + \sin^2\theta = 1.$$

Proof

*Proof.* The point $(\cos\theta, \sin\theta)$ lies on the unit circle (Definition 0.1), so its coordinates satisfy $x_1^2 + x_2^2 = 1$. □

**Example 0.7** (The unit circle and the angle $\pi/3$)**.** Take $\theta = \pi/3$, one sixth of the way round the circle. Figure 1 shows the point reached, $(\cos\theta, \sin\theta)$, and its two coordinates. The point is a vertex of a regular hexagon inscribed in the unit circle, whose sides have the same length as its radius, so the point is at distance 1 from $(1, 0)$. Writing $(x_1, x_2) = (\cos\theta, \sin\theta)$,

$$\begin{aligned} 1 &= (x_1 - 1)^2 + x_2^2 \\ &= x_1^2 + x_2^2 - 2x_1 + 1 \\ &= 2 - 2x_1, \end{aligned}$$

so $\cos(\pi/3) = x_1 = \frac{1}{2}$, and then $\sin(\pi/3) = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$. To three decimal places these are 0.5 and 0.866. As Theorem 0.1 says, the squared coordinates add to 1.

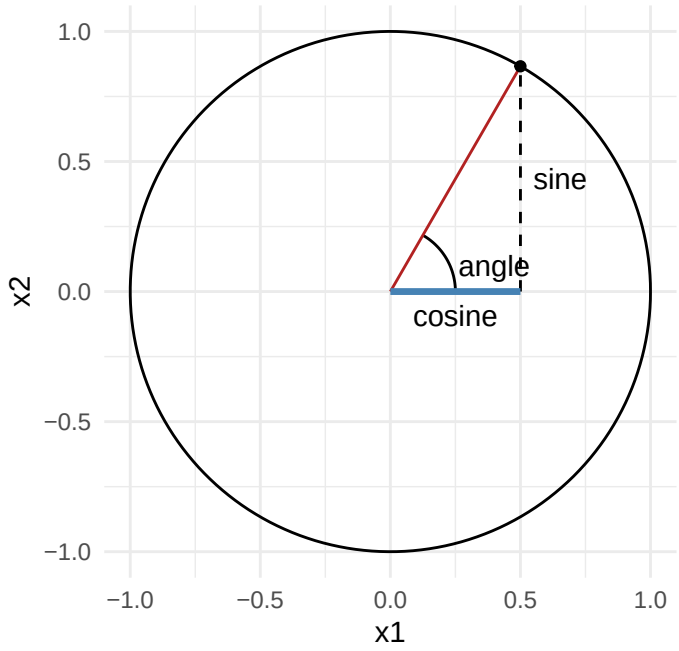



Figure 1: The unit circle with the point at angle $\pi/3$. Its horizontal coordinate is the cosine and its vertical coordinate is the sine.

**Theorem 0.2** (Cosine takes each value from $-1$ to 1 exactly once between 0 and $\pi$)**.** *The function* $\cos : [0, \pi] \to [-1, 1]$ *is decreasing, with* $\cos 0 = 1$ *and* $\cos \pi = -1$*, and takes each value in* $[-1, 1]$ *exactly once. On* $[0, \pi]$*,* $\sin \theta \geq 0$*.*

Proof

*Proof.* For $\theta \in [0, \pi]$, the point $(\cos\theta, \sin\theta)$ is at distance $\theta$ along the circle from $(1, 0)$, counterclockwise, so it lies on the upper half of the unit circle, $\{(x_1, x_2) : x_1^2 + x_2^2 = 1,\ x_2 \geq 0\}$, which gives $\sin\theta \geq 0$. Different values of $\theta$ in $[0, \pi]$ are different distances along that half circle, so they give different points. Two different points of the upper half have different first coordinates, because $x_2 = \sqrt{1 - x_1^2}$ is determined by $x_1$. So cos is one-to-one on $[0, \pi]$. Every $x_1 \in [-1, 1]$ is the first coordinate of the point $(x_1, \sqrt{1 - x_1^2})$ of the upper half, and the distance along the half circle from $(1, 0)$ to that point is some $\theta \in [0, \pi]$, so cos takes every value in $[-1, 1]$. Finally, move along the upper half circle from $(1, 0)$ to $(-1, 0)$ as $\theta$ increases from 0 to $\pi$. The first coordinate never takes the same value twice, so it never turns back. It starts at $\cos 0 = 1$ and ends at $\cos \pi = -1$, so it only decreases. □

**Definition 0.7** (Arccosine)**.** The **arccosine** $\arccos : [-1, 1] \to [0, \pi]$ is the function that sends each cosine, $c \in [-1, 1]$, to the one real number $\theta \in [0, \pi]$ whose cosine is $c$. By Theorem 0.2, there is just one such $\theta$.

**Example 0.8** (Three arccosines)**.**

- The arccosine of one is $\arccos 1 = 0$, because the cosine of zero is $\cos 0 = 1$.
- The arccosine of zero is $\arccos 0 = \pi/2$, because the cosine of a quarter turn is $\cos(\pi/2) = 0$.
- The arccosine of minus one is $\arccos(-1) = \pi$, because the cosine of a half turn is $\cos \pi = -1$.


[1] Wikipedia contributors, "Unit circle — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Unit_circle

[2] Wikipedia contributors, "Pi — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Pi

[3] Wikipedia contributors, "Sine and cosine — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Sine_and_cosine

[4] Wikipedia contributors, "Trigonometric functions — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Trigonometric_functions

[5] Wikipedia contributors, "Pythagorean trigonometric identity — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Pythagorean_trigonometric_identity